A First Course in Abstract Algebra

Section 47: Gaussian Integers and Multiplicative Norms

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

Contents

47 Gaussian Integers and Multiplicative Norms 3

Exercise 47.1: Splitting five 3

Exercise 47.2: Seven stays irreducible 3

Exercise 47.3: Factoring a norm twenty-five element 3

Exercise 47.4: Factoring a norm eighty-five element 3

Exercise 47.5: Failure of unique factorization 3

Exercise 47.6: One Gaussian division 3

Exercise 47.7: A Gaussian gcd 3

Exercise 47.8: Ten statements about norms 4

Exercise 47.9: Smallest nonunit norm 4

Exercise 47.10: Which integer primes split 4

Exercise 47.11: The Gaussian norm identity 4

Exercise 47.12: A multiplicative norm with square root minus five 5

Exercise 47.13: Existence of irreducible factorizations from a norm 5

Exercise 47.14: Another Gaussian gcd 5

Exercise 47.15: Finite Gaussian residue rings 5

Exercise 47.16: Imaginary quadratic norms 5
Exercise 47.17: Real quadratic norms 6
Exercise 47.18: Euclidean division with square root minus two 6

47 Gaussian Integers and Multiplicative Norms

Exercise 47.1: Splitting five

In $\mathbb{Z}[i]$, $5 = (2+i)(2-i)$. Both factors have norm 5, a rational prime, so a factorization of either would force one factor to have norm 1 and hence be a unit. Both are irreducible.

Exercise 47.2: Seven stays irreducible

The norm of 7 is 49. If $7 = \alpha\beta$ with both factors nonunits, their norms must both be 7. But $a^2 + b^2 = 7$ has no integer solutions: the possible squares at most 7 are 0,1,4 and none sum to 7. Thus such a factorization is impossible, and 7 is irreducible in $\mathbb{Z}[i]$.

Exercise 47.3: Factoring a norm twenty-five element

A direct multiplication gives $4 + 3i = i(2-i)^2$. Indeed $(2-i)^2 = 3 - 4i$ and multiplying by i gives $4 + 3i$. The factor i is a unit; $N(2-i) = 5$ is prime, so $2 - i$ is irreducible.

Exercise 47.4: Factoring a norm eighty-five element

We have $6 - 7i = (2+i)(1-4i)$. Expanding gives $2 - 8i + i - 4i^2 = 6 - 7i$. The norms are 5 and 17, respectively, both rational primes, so both factors are irreducible.

Exercise 47.5: Failure of unique factorization

Put $s = \sqrt{-5}$. Then $6 = 2 \cdot 3 = (1+s)(1-s)$. The norm $N(a+bs) = a^2 + 5b^2$ is multiplicative, and its units have norm 1. There are no elements of norm 2 or 3. Thus 2 (norm 4), 3 (norm 9), and $1 \pm s$ (norm 6) cannot split into two nonunits: their proper possible norm factors would require norm 2 or 3. All four displayed factors are irreducible. Their norms $4, 9, 6, 6$ show neither 2 nor 3 is associate to either $1 + s$ or $1 - s$. Hence the two factorizations are different up to associates.

Exercise 47.6: One Gaussian division

Compute

$$\frac{7+2i}{3-4i} = \frac{(7+2i)(3+4i)}{25} = \frac{13+34i}{25}.$$

Round both coordinates to nearest integers to choose $\sigma = 1 + i$. Then $(3-4i)(1+i) = 7 - i$, so $\rho = 3i$. We have $7 + 2i = (3-4i)(1+i) + 3i$, and $N(3i) = 9 < 25 = N(3-4i)$.

Exercise 47.7: A Gaussian gcd

The divisions

$$8 + 6i = i(5 - 15i) + (-7 + i), \qquad 5 - 15i = (-1 + 2i)(-7 + i)$$

have first remainder norm 50, smaller than 250. The next remainder is zero. Thus the last nonzero remainder $-7 + i$ is a gcd. Its associates, for example $7 - i$, are equally valid.

Exercise 47.8: Ten statements about norms

(a) True. The Gaussian integer norm makes the ring Euclidean, hence a PID.

(b) True. Round real and imaginary coordinates in Gaussian division.

(c) True: $n = n + 0i$.

(d) False: $1/2$ has no integer real coordinate.

(e) True, since the nonnegative remainder norms decrease.

(f) True. Norm divisibility restricts possible factor norms, as in Exercises 47.1–47.4.

(g) True. From $N(u)N(u^{-1}) = N(1) = 1$, integer norms give $|N(u)| = 1$.

(h) False. Degrees add rather than multiply; also the norm of 1 would be 0 rather than 1.

(i) True. For nonzero polynomials, degree addition gives $2^{\deg(fg)} = 2^{\deg f}2^{\deg g}$; declaring $N(0) = 0$ handles zero products and $N(1) = 1$.

(j) True. It is a subring of $\mathbb{C}$, hence a domain; Exercise 47.5 gives nonunique factorization.

Exercise 47.9: Smallest nonunit norm

Suppose $\pi = ab$ with both factors nonunits. Since $\pi \neq 0$ and the ring is a domain, $a, b \neq 0$. By the hypothesis relating norm and units, both $|N(a)|, |N(b)|$ are integers greater than 1. Minimality gives $|N(a)| \geq |N(\pi)|$. Multiplicativity then gives $|N(\pi)| = |N(a)||N(b)| > |N(a)| \geq |N(\pi)|$, a contradiction. Thus every factorization has a unit factor, proving irreducibility.

Exercise 47.10: Which integer primes split

(a) $2 = -i(1 + i)^2$, and $1 + i$ is irreducible because its norm is the prime 2.

(b) If $p \equiv 1 \pmod 4$, the sum-of-two-squares theorem gives $p = a^2 + b^2 = (a + bi)(a - bi)$ with both factors nonunits. If an odd prime $p \equiv 3 \pmod 4$ were reducible, both nonunit factor norms would equal p. But an integer square is 0 or 1 modulo 4, so $a^2 + b^2$ cannot be 3 modulo 4. This excludes norm p and proves irreducibility. Every odd prime is 1 or 3 modulo 4, completing both directions.

Exercise 47.11: The Gaussian norm identity

For $\alpha = a + bi$, $\beta = c + di$, multiplication gives $\alpha\beta = (ac - bd) + (ad + bc)i$. Therefore

$$\begin{aligned} N(\alpha\beta) &= (ac - bd)^2 + (ad + bc)^2 \\ &= a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 \\ &= (a^2 + b^2)(c^2 + d^2) = N(\alpha)N(\beta). \end{aligned}$$

The cross terms cancel. Since $N(\alpha) = a^2 + b^2$ is a sum of squares, it is nonnegative. Also $N(\alpha) = 0$ exactly for $a = b = 0$, and $N(1) = 1$, giving the stated norm properties.

Exercise 47.12: A multiplicative norm with square root minus five

Write $\alpha = a + b\sqrt{-5}$ and $\beta = c + d\sqrt{-5}$. Their product is $(ac - 5bd) + (ad + bc)\sqrt{-5}$. Expanding gives

$$(ac - 5bd)^2 + 5(ad + bc)^2 = (a^2 + 5b^2)(c^2 + 5d^2).$$

Thus $N(\alpha\beta) = N(\alpha)N(\beta)$. The values are nonnegative integers, $N(0) = 0$, and $N(1) = 1$.

Exercise 47.13: Existence of irreducible factorizations from a norm

Use strong induction on the positive integer $m = |N(a)| > 1$. If a is irreducible we are done. Otherwise $a = bc$ with b, c nonunits, necessarily nonzero. Their absolute norms are integers greater than 1 and multiply to m, so both are strictly smaller than m. By induction both factors are products of irreducibles; concatenate these factorizations. The smallest possible nonunit norm supplies the base case, since it cannot split as a product of two smaller norms greater than 1.

Exercise 47.14: Another Gaussian gcd

Gaussian division yields

$$16 + 7i = (1 + i)(10 - 5i) + (1 + 2i), \qquad 10 - 5i = (-5i)(1 + 2i).$$

The first remainder has norm 5, below $N(10 - 5i) = 125$, and the second remainder is zero. Thus $1 + 2i$ is a gcd, up to multiplication by $1, -1, i, -i$.

Exercise 47.15: Finite Gaussian residue rings

(a) Division by nonzero α gives each residue class a representative r with $N(r) < N(\alpha)$. There are finitely many integer lattice points in this bounded disk, so there are finitely many residue classes. More exactly the quotient has $N(\alpha)$ elements: multiplication by $a+bi$ has integer lattice matrix $\left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$ of determinant $a^2 + b^2$.

(b) An irreducible in this PID is prime, so its principal ideal is maximal. Its quotient is therefore a field.

(c) For (3), there are $N(3) = 9$ residues and characteristic 3. For $(1 + i)$, there are 2 residues and characteristic 2 because $2 = -i(1 + i)^2$. For $(1 + 2i)$, there are 5 residues and characteristic 5, since $5 = (1+2i)(1-2i)$. These are fields because the indicated divisors are Gaussian irreducibles (7-like reasoning for 3; prime norm for the other two).

Exercise 47.16: Imaginary quadratic norms

Let $s = \sqrt{-n}$, with n positive and square free.

(a) Multiplication and expansion give $N((a + bs)(c + ds)) = (ac - nbd)^2 + n(ad + bc)^2 = (a^2 + nb^2)(c^2 + nd^2)$. Values are integers, with $N(1) = 1$ and $N(\alpha) = 0$ only for $\alpha = 0$.

(b) If α is a unit, multiplicativity forces $N(\alpha) = 1$. Conversely, if $N(\alpha) = 1$, then $\alpha\bar{\alpha} = 1$, and $\bar{\alpha} = a - bs$ is in the same ring, giving an inverse.

(c) Every nonzero nonunit has integer norm greater than 1. Apply the strong-induction proof of Exercise 47.13 to obtain a factorization into irreducibles. This proves existence, not uniqueness.

Exercise 47.17: Real quadratic norms

For the intended real quadratic case assume the square-free integer $n > 1$. The representation $a+b\sqrt{n}$ is then unique because $\sqrt{n}$ is irrational. Define $N(a + b\sqrt{n}) = a^2 - nb^2$, allowing negative values.

(a) Direct expansion gives $(ac+nbd)^2-n(ad+bc)^2 = (a^2-nb^2)(c^2-nd^2)$, so the norm is multiplicative. A nonzero element has nonzero norm by irrationality of $\sqrt{n}$.

(b) The appropriate statement is $|N(\alpha)| = 1$ if and only if α is a unit. A unit has integer norm dividing 1. Conversely, for norm ± 1, $\alpha^{-1} = \bar{\alpha}/N(\alpha)$ lies in the ring. Requiring $N = 1$ literally would be false, e.g. $1 + \sqrt{2}$ has norm -1 and inverse $\sqrt{2} - 1$.

(c) The absolute norm is a positive integer, greater than 1 on nonzero nonunits. The induction in Exercise 47.13 therefore proves existence of an irreducible factorization.

The original reference includes square-free $n = 1$; here the expression $a^2 - b^2$ is not well defined on $a + b\sqrt{1}$ (1 has representations $(1, 0)$ and $(0, 1)$). Thus that degenerate case must be excluded for this norm construction.

Exercise 47.18: Euclidean division with square root minus two

For nonzero β, write $\alpha/\beta = u + v\sqrt{-2}$ with $u, v \in \mathbb{Q}$. Choose integers m, k with $|u - m| \leq 1/2$ and $|v - k| \leq 1/2$. Put $q = m + k\sqrt{-2}$ and $r = \alpha - \beta q$. Then

$$N(r) = N(\beta)\big((u - m)^2 + 2(v - k)^2\big) \leq \tfrac{3}{4}N(\beta) < N(\beta).$$

If $r = 0$ the required alternative holds as well. All nonzero norms are positive integers and multiplication multiplies them by an integer at least 1, so the second Euclidean axiom holds too. Hence $\mathbb{Z}[\sqrt{-2}]$ is Euclidean.